MacFast

Department of Chemistry and Chemical Biology
Department of Kinesiology
McMaster University

What ?

- MacFAST is a scalable web-based platform for practice testing using Computerized Adaptive Testing (CAT), designed to support effective studying and long-term retention of material.
- This approach also aligns with the preferred study strategies of students in introductory chemistry and kinesiology courses, based on internal research by Dr. Greenberg.

Challenges

- How can adaptive testing work as a study tool?
  - Adaptive testing is usually used to measure ability, not support studying. MacFAST uses ability estimates to avoid giving questions that are too difficult or not yet appropriate for the student.
- How can instructors use it easily from the start?
  - MacFAST supports question uploads in common department formats, like CSV & Word documents.

Why ?

- Adaptive testing uses statistical data to give each student individualized questions: correct answers lead to harder questions, while incorrect answers lead to easier ones.
- This creates a personalized practice experience that improves exam preparation, builds metacognitive skills, and keeps students challenged at the right level while giving them a realistic view of their progress.

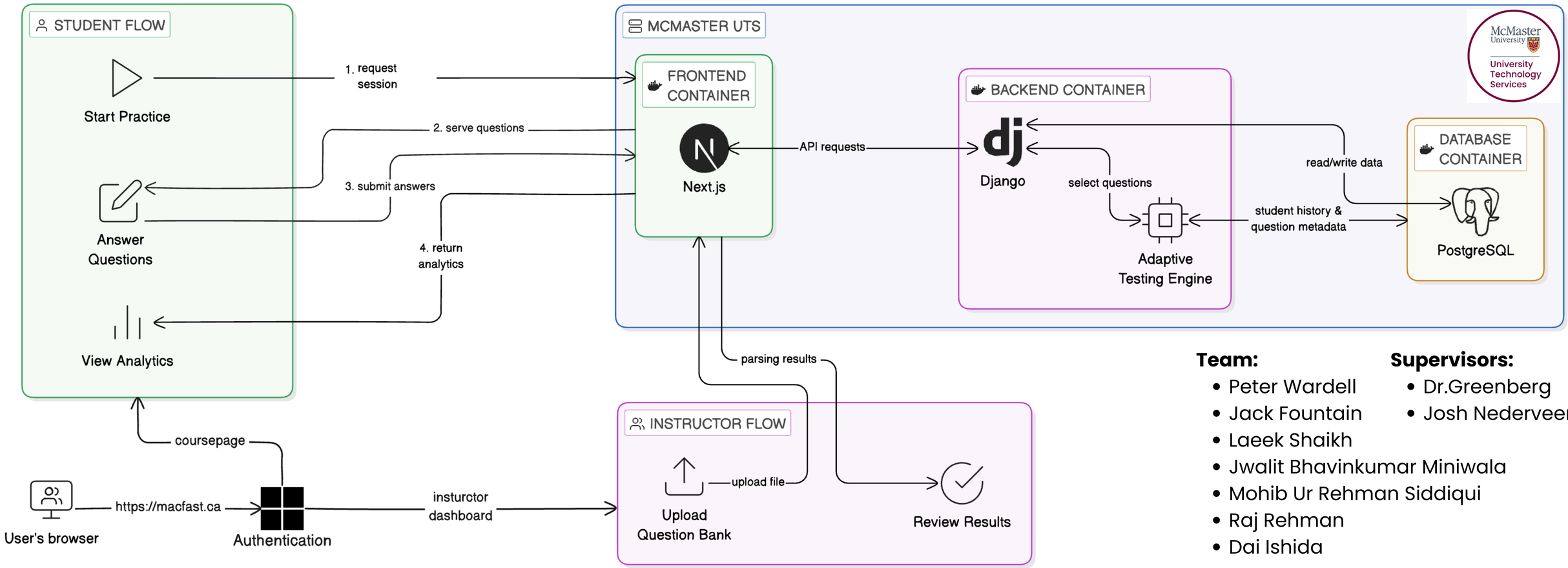
How does it compare?

Consideration	Online HW	Avenue Quizzes	Problem Sets	Mac-FAST
Content	✗	✓	✓	✓
Question Difficulty	✗	✓	✓	✓
Technical support	✓	✗	✗	✓
Cost to students	✗	✓	✓	✓
Multiple attempts	✓	✓	✗	✓
Feedback & hints	✓	✗	✗	✓
Adaptive	✗	✗	✗	✓

Key: ✓ = Excellent Performance ✗ = Limitation ▭ = Mac-FAST (Highlighted Premium Solution)

How?

- MacFAST uses a Next.js frontend that communicates with a Django API over HTTPS, with both services running as Docker containers on McMaster's UTS virtual machines. Authentication is handled through McMaster SSO using OAuth 2.0 / OIDC.
- All data is stored in a Postgres database. The adaptive logic in the Django backend uses this data to select questions and return responses to the Next.js frontend, which then delivers the interface to students and instructors.
- MacFAST uses the Rasch Model (1PL) for adaptive testing. This model assigns a difficulty value to each question and estimates each user's ability. As users submit responses, the system updates their estimated score to present questions at an appropriate level. Once the estimate reaches an acceptable level of accuracy, the user is prompted to move on to a different subtopic or unit.



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